fn gamma_hi_from_x

Michael Shoemate

April 22, 2026

Proves soundness of `fn gamma_hi_from_x` in `mod.rs`.

1 Hoare Triple

Precondition

`x` is finite.

Pseudocode

```
1 def gamma_hi_from_x(x: float):
2     expm1_lo = (-x).neg_inf_exp_m1()  #
3     return -expm1_lo
```

Postcondition

Theorem 1.1. For any setting of the input parameter `x` such that the given precondition holds, `gamma_hi_from_x` either returns `Err(e)`, or returns `Ok(out)` where

$$\texttt{out} \geq 1 - e^{-\texttt{x}}.$$

2 Proof

Assume the precondition is met.

Lemma 2.1. `gamma_hi_from_x` only returns `Err(e)` if the call to `neg_inf_exp_m1` on line 2 returns `Err(e)`.

Proof. The only fallible operation in `gamma_hi_from_x` is the invocation of `(-x).neg_inf_exp_m1()` on line 2. All other operations are infallible. □

Lemma 2.2. If the outcome of `gamma_hi_from_x` is `Ok(out)`, then

$$\texttt{out} \geq 1 - e^{-\texttt{x}}.$$

Proof. By the proof definition of `neg_inf_exp_m1`,

$$\texttt{expm1_lo} \leq e^{-\texttt{x}} - 1.$$

Multiplying by -1 reverses the inequality:

$$-\texttt{expm1_lo} \geq 1 - e^{-\texttt{x}}.$$

Since the function returns $\texttt{out} = -\texttt{expm1_lo}$, the claim follows. □

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.2. □